



Abacus

c. 2700 BCE
First abacus

The abacus is invented in Sumer (present-day Iraq) and is soon widespread.

1617

Napier's Bones

Invented by John Napier, this set of inscribed rods or "bones" provides a quick way of multiplying or dividing large numbers.



Napier's Bones

1642

Pascaline

Blaise Pascal invents one of the first mechanical calculators capable of simple arithmetic.



Pascaline



Arithmometer

1820

Arithmometer

Thomas de Colmar builds the first commercially successful mechanical calculating machine.

c. 100 BCE
Antikythera mechanism

This early Greek device uses bronze gears to calculate astronomical positions.



Antikythera mechanism

1630

Slide rule

Invented by English mathematician William Oughtred, the slide rule is used to multiply, divide, and calculate roots and logarithms.

Slide rule

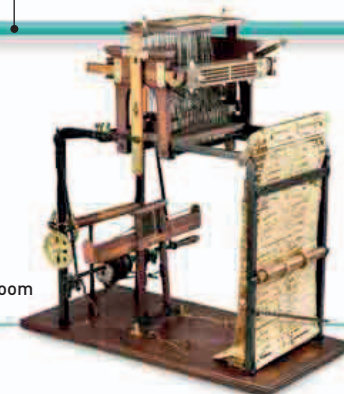


1801

Mechanical loom

Joseph Marie Jacquard's loom is controlled by cards with punched holes, a system later adopted by early computers.

Jacquard loom



THE STORY OF CALCULATING MACHINES

CALCULATING HAS BEEN IMPORTANT FOR SCIENCE, INDUSTRY, AND COMMERCE FROM THE EARLIEST TIMES TO THE PRESENT DAY

The word *calculate* is derived from the Latin *calculus*, or "little stone," referring to the ancient practice of using stones to perform calculations. Since then, increasingly sophisticated devices have been invented to perform the complex calculations demanded by advances in the sciences.

The first calculating device, the abacus, evolved when counting stones were arranged on a frame, and remained the most widely used means of calculating until the 17th century. A breakthrough came with Scottish mathematician John Napier's discovery of logarithms (see 1614–17), and his invention of the calculating device known as Napier's Bones. The first mechanical calculators also appeared in the 17th century, prompted by the need for accurate astronomical tables.

PROGRAMMABLE MACHINES

During the Industrial Revolution, a French weaver, Joseph Marie Jacquard, used punched

cards to control the working of his looms. The idea of a calculating machine capable of carrying out different programmable functions was originated by British mathematician Ada Lovelace. The same idea was taken up by British inventor Charles Babbage in his concept of an "Analytical Engine."

The first electrical computers began to appear in the 1930s, and with the introduction of integrated circuits, or chips, smaller and more powerful computers and calculators eventually became viable. By the mid-1970s, entire processing units on silicon chips—microprocessors—enabled the production of personal computers.



BINARY NUMBERS

Unlike the decimal system, which uses the numbers 0 to 9, the binary numeral system uses only two symbols, 0 and 1. In this system, 1 is represented by 1, 2 by 10, 3 by 11, 4 by 100, and so on. Because there are only two symbols, the binary system is ideal for use in digital electronic computers, where the two possible states of an electronic circuit, off or on, correspond to the digits 0 and 1.

“ALL WHICH IS **BEAUTIFUL**
AND **NOBLE** IS THE RESULT OF
REASON AND **CALCULATION.**”

Charles Baudelaire, French poet (1821–67)